ABV IIITM STUDENT BRANCH

IEEE MP SECTION

HACKSAGON

A NATIONAL LEVEL SOFTWARE & HARDWARE HACKATHON

RFID - Integrated Child Safety & Attendance System

Neural Nexus

THEME OF THE PROJECT

This project involves both software and hardware applications

Hardware Used:

- ESP32 (Scanning Bench & Live Tracker)
- RFID Module (RC522)
- NEO-6M GPS Module
- Li-Ion 2500mAh Battery + TP4056 + MT3608
- LEDs & Buzzer for alerts

Software Used:

- ESP-NOW (Wireless Communication)
- Web Dashboard (for location tracking)
- Arduino IDE (Coding & Flashing)
- Firebase (Database for real-time data storage)

PROBLEM STATEMENT

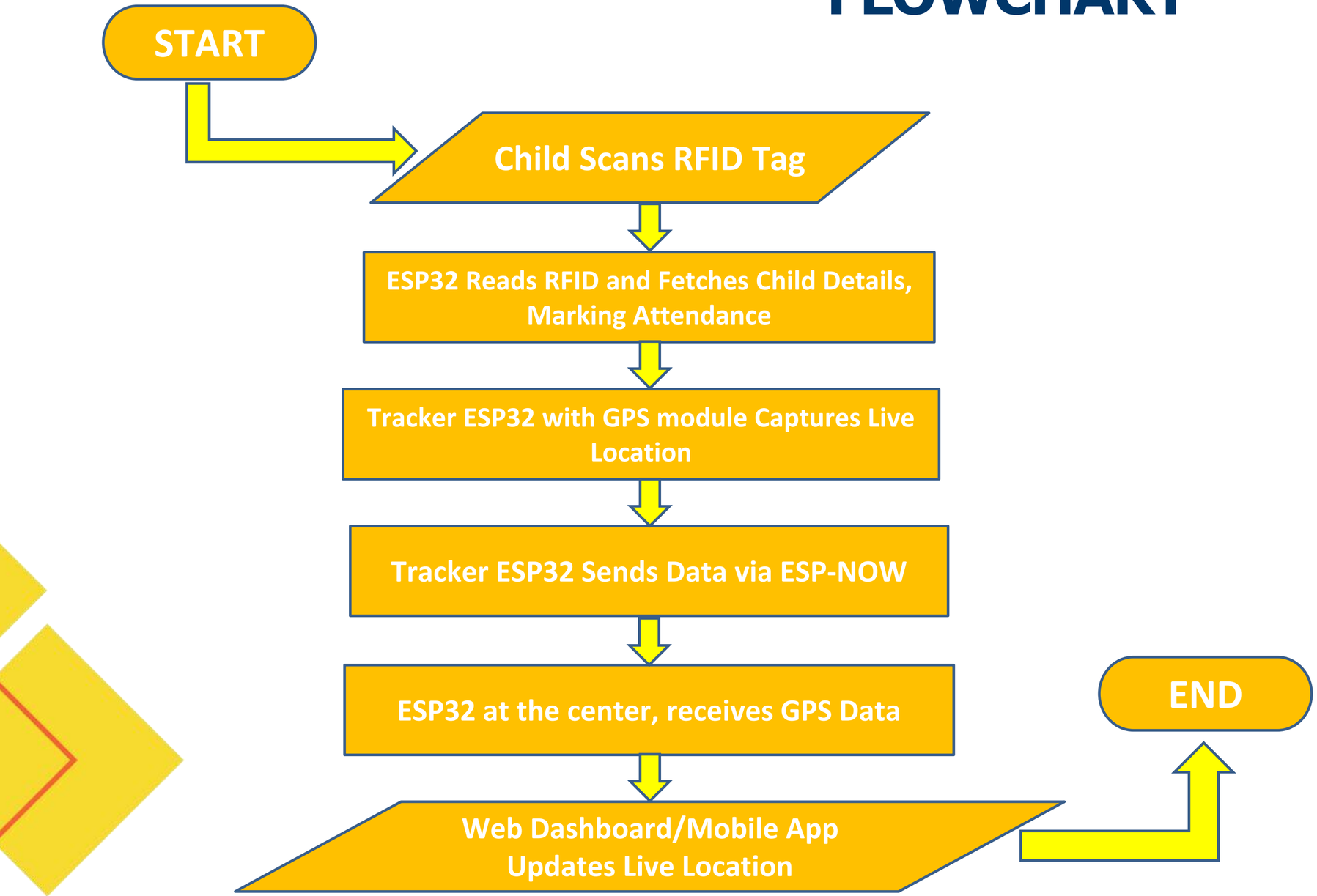
- Ensuring child safety during school commutes is a major concern.
- Traditional attendance systems lack real-time tracking of children.
- Parents and school authorities struggle to monitor live location.
- No direct, cost-effective, and efficient solution exists to track children within a specific area.

PROPOSED SOLUTION

Our system consists of four major parts:

- **Scanning Bench** – Uses an ESP32 with an RFID module to scan the child's ID card, log attendance, and store details.
- **Live Tracker** – A GPS-enabled ESP32 device placed in the child's ID card, transmitting real-time location.
- **Display Setup** – Receives location data via ESP-NOW and displays it on a web dashboard for easy monitoring.
- **Mobile App** – Records and Updates child live location on the regular basis.

FLOWCHART


START
Child Scans RFID Tag
ESP32 Reads RFID and Fetches Child Details, Marking Attendance
Tracker ESP32 with GPS module Captures Live Location
Tracker ESP32 Sends Data via ESP-NOW
ESP32 at the center, receives GPS Data
Web Dashboard/Mobile App Updates Live Location
END

FLOWCHART DESCRIPTION

- **RFID Attendance System**

  Child scans RFID tag on an ESP32-based scanning bench.
  Marks attendance & stores child's details
  (Name, Contact, Parents' Info, Address).

- **Live GPS Tracking**

  Child carries an ESP32-based GPS tracker
  (GPS module + Battery Module).
  Live coordinates sent via ESP-NOW to another ESP32.

- **Data Storage & Dashboard**

  GPS data stored in Firebase.
  Real-time location displayed on a web dashboard/Mobile App.

FEATURES AND NOVELTY

- **Live Tracking** – Real-time GPS-based location tracking of children.
- **Wireless Communication** – Uses ESP-NOW instead of Wi-Fi/Bluetooth for efficient data transmission.
- **Automated Attendance** – RFID-based scanning logs attendance instantly.
- **Web Dashboard** – Provides easy monitoring of child's location.
- **Low-Cost Solution** – Uses affordable hardware compared to commercial trackers.

DRAWBACK AND SHOWSTOPPERS

- **Power Management Challenges** – Tracker needs an optimized battery solution to last the entire day.
- **Obstruction Issues** – GPS accuracy may be affected by buildings or tunnels.
- **Data Security Concerns** – Requires secure transmission to prevent unauthorized access.

Neural Nexus

PRATIK TIWARI

Contact:
Email:
24tiwarip@rbunagpur.in
Phone: 9096594439

VEDANT SIKARIYA

Contact:
Email:
24sikariyav@rbunagpur.in
Phone: 7709717633

VIDHI UMAP

Contact:
Email:
24tumapv@rbunagpur.in
Phone: 9373112458

THANK YOU